

Chapter 3: Model and Simulation

In computational lithography, patterning operations, i.e. **exposure** of photoresist to DUV or EUV lights propagating from source through mask and projection optics to resist plane and **post exposure bake/development/etch** of the resist, are described numerically by **Optical Model** and **Resist/Etch Model**, respectively. Using these models one can predict how the geometry and the context of a mask pattern influence its printing on wafer.

Industrial-grade models are usually labeled as “rigorous” or “compact.”

Rigorous modeling calculates light propagation, interference, diffraction, and interaction with resist from “first principles,” i.e. Maxwell equations of electromagnetism and kinetic theories of photon absorption, acid diffusion and reaction with polymer, dissolution by developer and erosion by etchant. Such modeling may be highly accurate and yet extremely computationally expensive. Well-known brands of rigorous modelings include S-Litho, Prolitho.

Compact modeling approximates light field in resist and resist responses to light, heat, developer, and etchant with various “kernel” functions. Light field is computed with Hopkins imaging theory and Sum of Coherence Approximation. Resist response is derived from **functions of light field**, including intensity and its derivatives, density, convolutions, etc. Resist/Etch models contain parameters that need to be **calibrated**, i.e., derived from empirical data. Well-known brands of compact modelings include Calibre, Proteus, Tachyon.

Only compact models can be made accurate and efficient enough for **full-chip** RET. We shall focus on compact models in this chapter.

(Recent development in **mask error modelling** of how mask writing tool generating mask errors, including bias, corner rounding, caused by density variation, and electron bur, resist diffusion, etc., will be covered in Advanced Topics).

Section One: Optical Model

An optical model is based on **Hopkins imaging theory** of partially coherent light. Here are key concepts of the theory:

Partial Coherence: Light source in lithography is composed of incoherent, quasi-monochromatic, continued source elements.

With such source, in the field of light propagation, for two points with the space between them much smaller than “**coherence length**” of one single source point (Born, p 507), the interferences of lights propagating from these two points can be characterized by “mutual intensity,” (Born, p509), which is the most important property of **partially coherent light field**.

Optical Model

Transmission Cross Coefficients (TCC): With Hopkins theory, image of an **object**, in our case, a pattern on photomask, is defined as **convolution** of the object with **Transmission Cross Coefficients** (Born, 513, 531). TCC is derived from the source and the pupil function of the optical system, in our case, the objection lens.

Therefore, optical modeling is fundamentally TCC computation. Convolution of optical model with mask pattern produces **Aerial Image**, which is light intensity on focus plane as if NO photoresist exists. How such light interact with photoresist will be described in Resist Model.

An optical model is built with predetermined parameters of the optical system, including source, lens, and mask. No empirical data is needed, although some optical modeling method allow slight correction of predetermined parameters to compensate for deviation of the real system from the predetermined values.

Optical Model

Sum Of Coherence (SOC) Approximation: TCC is usually calculated in frequency space. It is a **four-dimensional integral** of Fourier Transform of source and pupil over the “effective domain” of spectrum (Born 531). Such calculation is highly computationally expensive.

One major approximation in Compact Modeling is based on SOC (Cobb). TCC can be expressed as sum of **two-dimensional integrals** over coherent frequencies, kernels, which are mostly `bessel_sinc` functions. Number of kernels determines accuracy of the approximation.

An optical model is a directory, which contains the kernels (usually in binary forms) and text files showing the parameters. User can define the number of retained kernels. A kernel has an “**ambit**,” which is the radius of the sphere of influence of the kernel. Mask patterns are at the **proximity** of each other if they are less than the ambit apart.

Optical Model

Due to the proximity effect, two **identical** shapes on a mask may have **different** images, if their **environments**, i.e., other shapes within the ambit, are different. In litho simulations, the environment is often referred to as **context**. In circuit design, one device can be placed numerous times, each placement may have different context. Regardless of the context, a device must deliver same target performance. Therefore, proximity effect needs to be corrected through RET technologies, OPC, ILT, etc.

Proximity effect originated from Mutual Intensity of particle coherent light. The Mutual Intensity between two locations decreases fast with the spaces between them. For example, ambit can be estimated (wang, p170):

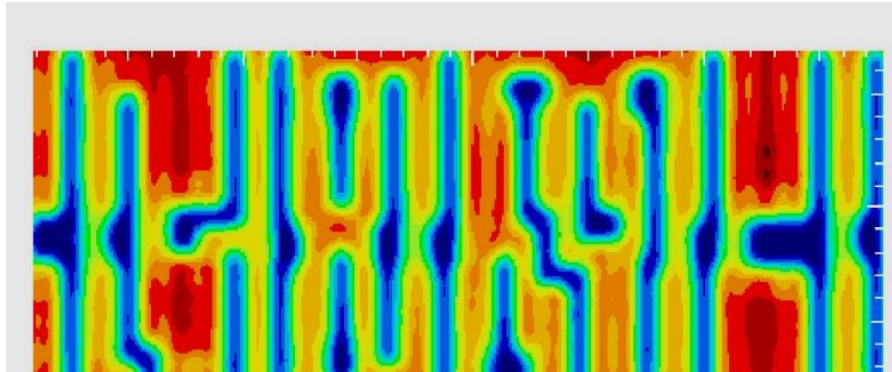
$$R_a = 0.61 * (\lambda / \sigma * NA)$$

Coherence length is defined as (Born, 507) $R_c = \lambda^2 / \Delta\lambda$. Take $\lambda = 193 \text{ nm}$, $\Delta\lambda = 10\% \text{ of } \lambda$, $\sigma = 0.5$, $NA = 0.75$. $R_a = 314 \text{ nm}$, $R_c = 1930 \text{ nm}$. Therefore, $R_a \ll R_c$. Partial coherence theory is valid for patterns within the ambit.

Simulation with Optical Model

As we shall learn later, reactions of resist to light field are modeled as functions of light field. Therefore **Aerial Image** simulations is the very foundation upon which other simulations can be conducted. As a rule of thumb, accuracy of litho simulations is 90% optics and 10% resist.

Since optical model can be created without fitting empirical data, it can be employed conveniently to provide qualitatively good assessment of imaging quality of mask patterns.



Metric of Imaging Quality

Imaging quality is commonly assessed with the following numeric or graphic indicators taken on test patterns and hotspots (to be describe later):

Image: A map of pixelated, colored intensity distribution over simulation area, as shown in previous slide. Intensity value at each pixel can be extracted.

Contour: A equal-intensity delineation on the image.

Slice: A plot of intensity or Normalized Image Log Slope (NILS) varying continuously along a linear gauge.

CD: The distance between two consecutive points on a gauge where intensity slice crosses the contour at threshold value.

MEEF: Place a gauge across a line and take CD. Vary the linewidth and take CD again.

$$\text{MEEF} = \Delta(\text{CD})/\Delta(\text{width}).$$

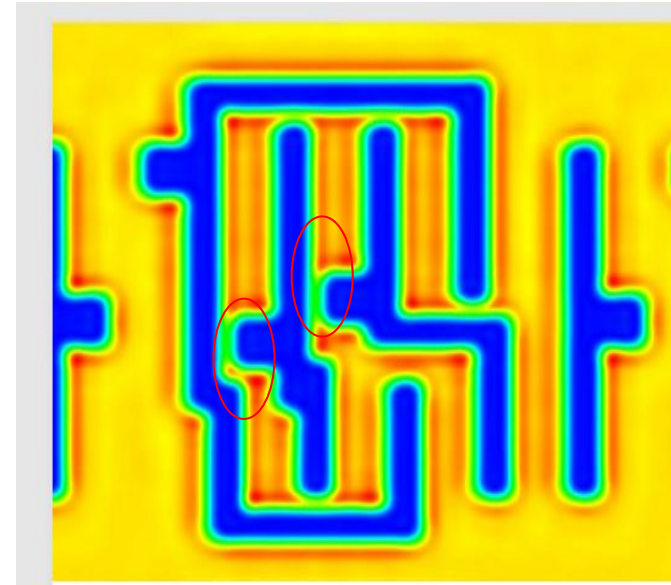
MEEF is a powerful indicator of image distortion. MEEF equals to one is ideal.

NILS: Place a gauge across a line of width w . At each point x of the gauge, $\text{NILS} = (w/l) \cdot (dl/dx)$. Larger NILS means sharper image contrast and is therefore desired.

Metric of Imaging Quality (continued)

Exposure/Focus Latitude: In a production environment, variation of exposure and focus from set values cannot be avoided, due to finite spec of optical system, wafer flatness (even with CMP), etc. Imaging quality must hold within the spec for Exposure/Focus latitude required by a fab. For example, CD variation should not exceed 10% with exposure variation of 10% and defocus variation of $\pm 100\text{nm}$, and hotspots should not fail to print, as illustrated in the snapshot. (p634, Handbook)

Note: All the above numerical/graphic indicators of imaging quality can also be obtained with a calibrated model.



These hotspots fail at over-exposure condition.

Light Field Inside Resist

With an optical model, aerial image can be calculated. However, in reality, the best focus plane is typically at $\frac{1}{3}$ of the resist height from the top of the resist. As light enters resist, which can consist of multiple layers of thin films stacking upon Si substrate, refraction/absorption/reflection must be considered.

Absorption: Light intensity decrease with the depth of resist as $I(x, y, z) = I(x, y, 0) \exp(-\alpha z)$, where α is absorption coefficient.

Standing Wave Effect: Light entering the resist and reflected from the resist-substrate interface would interfere to form standing wave, which manifests in the vertical profile of the resist.

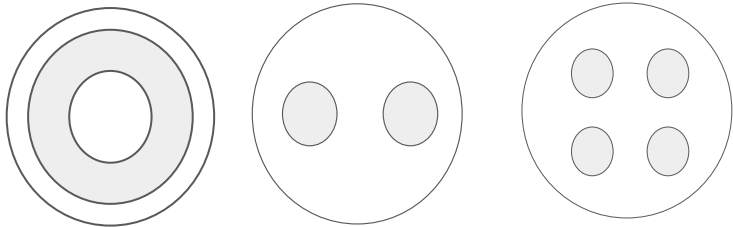


Scalar vs. Vector Optical Model

In the early days of litho simulations, electric field in both polarization directions was considered scalar. The assumption is a good approximation for near-normal entrance.

When light enters resist with large oblique angles—situations with large Numerical Aperture (NA), i.e., > 0.7 , and off-axis illumination—the electric field in the plane of entrance must be computed as vector.

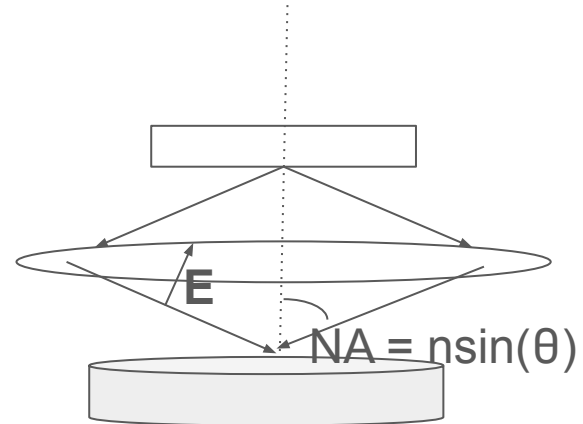
Examples of Off-Axis Illumination



Mask

Lens

Wafer



Building Optical Model on EDA Platform

As we have learned, Optical Models are based on well-established principle and theories. However, to create an optical model, each EDA platform employs unique and proprietary numerical representation, an engine, that features certain numerical methods, programming languages, architecture of algorithm, as well as spec and limitation for input and output. The insight of the engine is a black box to the users, for whom interactions with the engine can only be done via a GUI. In the GUI, the user defines the following parameters and launches the engine, usually with a single command. Optical model directories will then be generated.

Source: wavelength (λ), source shape, partial coherence (σ), intensity distribution

Lens: reduction ratio, aberration, apodization, NA, polarization, pupil shape

Mask: substrate (SiO₂), chrome or MoSi or multilayer reflector; Transmission coef, bias/corner rounding

Resist: n/k/thickness of TARC, resist, BARC, hard mask, and substrate; image depth; exposure/defocus

Other Inputs: scalar or vector Model; thin or thick mask; etc.

Section Two: Resist/Etch Model

As mentioned earlier, lithographic models can be used to predict how geometry and context of a mask pattern influence its printing on wafer, and such influences are 90% dictated by light field. The rest are captured by After Development Intensity (ADI) model, predicting how photoresist transforms when exposing to light, heat, and developer, and After Etch Intensity (AEI) model, predicting how unprotected wafer materials are etched away. We discuss ADI first.

Dynamical processes of photoresist transformation include absorption, diffusion, deprotection/protection, and dissolution. ADI models describes these processes as functions of light field. In Rigorous Modeling, these functions are based on kinetic theories. In Compact Modeling, the functions are approximated by local variations such as derivatives of image intensity, or by convolution of the light field with kernels characterizing spatial correlations—Variable Threshold Model and Equal Threshold Models, respectively, as we shall discuss later.

Resist physical/chemistry properties, i.e., diffusivity, absorption rate, reaction rate, are used explicitly in Rigorous Modeling, and implicitly in Compact Modeling in which they are embedded in coefficients of model terms. The coefficients are derived by fitting empirical measurements.

Physics/Chemistry of ADI Modeling

An ADI model describes how resist transforms during the following steps of patterning:

Exposure: Resist is exposed for ~ 2 second at room temperature to DUV/EUV lights. As light propagates in the resist, photons are absorbed by light-sensitive component, generating acid, and by other components. Concentration of the light sensitive component, M varies with time according to this equation:

$$\partial M / \partial t = C I(x, y, z) M$$

From which, at the end of exposure, a **spatial profile** of an **initial acid concentration, H** , as the function of light intensity $I(x, y, z)$ can be derived:

$$\text{Acid Concentration } H = 1 - M = 1 - \exp(-C * \text{Exposure Energy } (x, y, z))$$

As mentioned earlier on, $I(x, y, z) = I(x, y, 0) \exp(-az)$. The absorption coefficient a can be calculated from Dill's Theory and Dill coefficients. (Dill, 1974)

Physics/Chemistry of ADI Modeling (continued)

Post Exposure Bake (PEB): PEB of resist typically takes 60-120 seconds at elevated temperatures of 90 to 130 degrees. Heat facilitates acid diffusion, while acid triggers catalytic chain reactions to deprotect polymers. Diffusion-reaction model of acid concentration H , base quencher concentration Q , and protected polymer concentration W , can be found (Mack):

$$\partial H / \partial t = \nabla \cdot (D_H \nabla H) - K_{loss} H - K_{quench} H Q$$

$$\partial Q / \partial t = \nabla \cdot (D_Q \nabla Q) - K_{quench} H Q$$

$$\partial W / \partial t = -K_{amp} H W$$

In the first two equations, the first term comes from Fick's Second Law of Diffusion. These equations show how light intensity and its 1st and 2nd derivatives shape spatial profile of deprotected polymers, $(1 - W)$, that is the **latent image**.

Physics/Chemistry of ADI Modeling (continued)

Development: In the development step, huge difference in solubility of unprotected/protected polymers in chemical solutions, i.e., developer, transform the latent chemical image to 3D physical structure. **Development rate** as function of latent image, $(1-w)$, can be described by this equation (Mack):

$$R(w) = R_{\max} \frac{(a+1)(1-w)^n}{a + (1-w)^n} + R_{\min}$$

$$a = \frac{(n+1)}{(n-1)} (1 - m_{th})^n$$

ADI Model Forms

With development complete, the lithography loop closes:

Light Field -> Acid Profile -> Deprotection Image (Latent) -> Physical Topography (Resist Profile)

A Compact resist model predicts resist profile using functions of light field. Model functions can be in various forms. Two classes of mainstream model forms are:

Variable Threshold Method: At a location (x, y) of image plane, if light intensity **equals to a threshold value** then this location is **on the edge** of the resist profile. The threshold value is computed as **linear combination** of a constant, intensity slope and curvature, max/min of intensity, density, and other local properties. Therefore, the threshold **varies continuously** with **location**. If intensity is below or above the threshold, then this location is inside/outside (depending on resist tone.) **Coefficients** of linear combination must be derived empirically.

ADI Model Forms (continued)

Equal Threshold Method: Resist profile is a **contour of a field**. The field is composed of intensity plus **perturbation terms to the intensity**. These terms aim to capture physics/chemistry of absorption, diffusion, deprotection and development. In this example, the perturbations are described as convolution of initial acid profile with kernels for PEB and Development. Kernels can be Gauss, Top Hat, std_deviation, disk, quad, Zernike polynomial. For example, a Gauss function with radius ~ 40 nm, roughly the acid diffusion distance.

$$R(x,y) = C0 \cdot I(x,y) + C1 \cdot K_{\text{PEB}} \otimes (1 - e^{-(c \cdot I(x,y))}) + C2 \cdot K_{\text{develop}} \otimes (1 - e^{-(c \cdot I(x,y))})$$

A resist profile is a contour of points with **equal field value**, that is a **threshold that user defines**.

ADI Model Calibration

Once a **model form** is selected, **coefficients** in the form are then derived empirically.

For a uncalibrated resist model, resist profile is defined by light intensity at a **fixed threshold value**, usually around 0.3.

For a calibrated resist model, coefficients are determined by fitting process, in which the coefficients are allowed to vary within certain range and with certain precision. For each combination of the coefficients, a model is created. Resist profile of **test patterns simulated with** the model is compared against measured profile. By minimizing the difference between them without so called “over fitting,” i.e., generating results that do not make sense in reality while mathematically highly accurate for some test patterns, a calibrated model is selected.

The most important step in model calibration is the design of **test patterns**. We shall discuss test patterns in great details next.

Once test patterns are printed on wafer, resist CD, i.e., line width and space, and resist image will be collected with Scanning Electron Microscopy (SEM). Typically, CD would be used in **model fitting** and images would be used for **model verification**, although image-based model fitting is gaining interests. This topic will be touched in New Trends in RET.

Test Pattern and Measurement

Standard Test Patterns: As mentioned in Photomask Fundamental, mask patterns printed on wafer suffer from distortions that are characterized by **thru-pitch CD variation, corner rounding, line-end shortening**, etc. Standard test patterns is a collection of **simple** patterns, each, per simulation or wafer measurement, would capture ONE type of the above distortions quantitatively. Most common of them will be shown in next slides.

Hotspot Test Patterns: These are **complicated** and random patterns, which, based on experience, tend to print badly due to multiple types of distortions. Some examples would be shown later.

RET test patterns can be included in test chip for which a mask would be dedicated, or in **scribes** of a production mask, i.e., margins around chips wherein metrology patterns, electrical test patterns, etc, are located.

Once test patterns becomes part of mask layout, RET engineer must provide Patterning engineer a **datasheet**, i.e., an Excel file. The datasheet specifies locations of each individual test pattern and measurement type, i.e., line, space, image, etc. Patterning engineers fill the datasheet with wafer data/image. It is crucial for RET engineer to cross exam the data/image to identify error and inconsistency and to request clarification or re-measurements. Without such **clean-up**, the ensuing model fitting process may go as “garbage in garbage out.”

Generate Standard Test Patterns on EDA Platform

Each EDA platform has its own test pattern generation tool. Typically, user can launch such tool from UNIX command line, then enter the following geometry/dimension information to create test pattern **layout** and corresponding **datasheet**.

Pattern Type	Geometry	Dimension Variables	Measurement and Purpose
1D Line/Space	single or 1D array of lines; horizontal/vertical or rotated by any angle	Line width/space; Assist feature width/space/number	Measuring CD to capture lineality, thru-pitch variation and HV variation. Imaging line edge roughness.
1D Line End	single or 1D array of lines facing each other; horizontal/vertical or rotated by any angle	Line width/space; Gap between lines; Assist feature width/space/number	Measuring tip-to-tip space to capture line end shortening as function of line width/pitch and gap. Imaging line end profile.
2D Square/Rectangle	single or 1D chains or 2D arrays (aligned or staggered) of squares/rectangles	Width/Aspect Ratio; Space; Row shifts; Assist feature width/space/number	Measuring CD as function of width/space/shifts; imaging corner rounding.

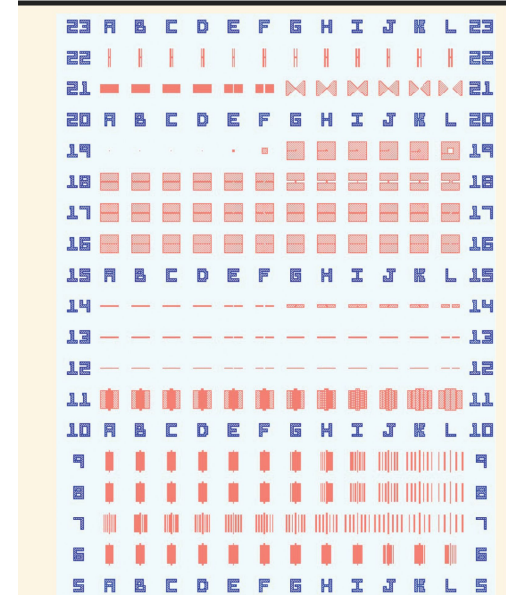
Layout of Standard Test Pattern

A layout generated by any EDA test pattern generator would contain shapes, numeric or text row/column indicators, and other metrology labels.

Usually, test pattern generator provides templates with default arrangements of patterns, i.e., how far should patterns be apart, how should they be ordered by row and column, where to place metrology labels and indicators, what layer/datatype/grid/ should be used for shapes, labels, and indicators, etc. The defaults are designed around technology node that the user specifies.

Test pattern generators would allow user to add any individual patterns by specifying type/geometry/dimensions in the previous slide, and/or provide some “wizards” that automatically select most common patterns to generate.

In this example, the user wants only 1D line/space/line end patterns. The user should specify a “nominal” CD that is consistent with technology node, say 45nm. The user would want to create line, space, line ends, etc, with dimensions ranging from 45nm to 55nm with 1nm step, and then from 55nm to 75nm with 5nm step. The idea is that patterns with dimensions close to nominal CD are most presenting on chips, most difficult to print well, and therefore need to be most accurately modeled, so 1nm step should be used.

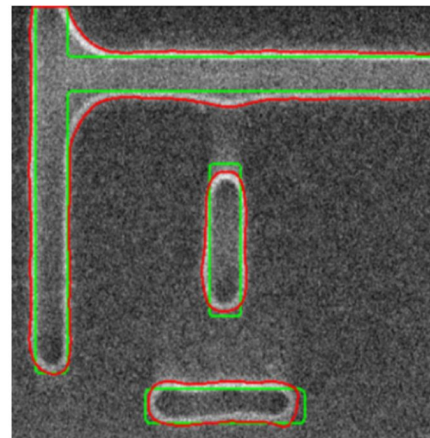
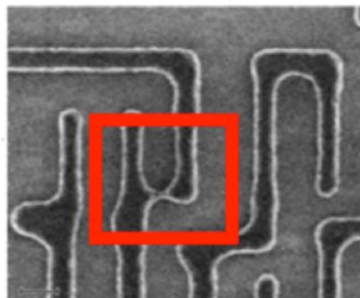
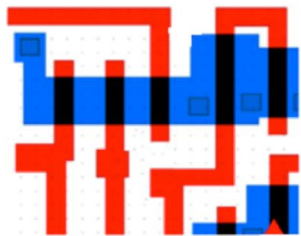
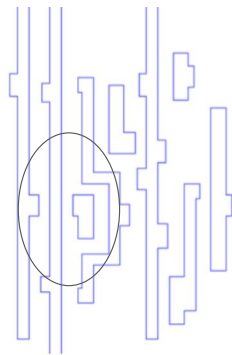


Hotspots

Hotspots are densely packed, complicated, random 2D patterns. In such a pattern, multiple nominal space/lines, corners and line ends entangle with each other. Resolving one nominal feature may conflict with resolving another.

Usually hotspots are discovered by yield analysis because they fail to resolve, as shown in the example. Recently, pattern analysis technology has become mainstream, with which hotspots can be located from design layout, or generalized and parameterized into special test patterns. For model calibration, SEM image around hotspots are taken and overlaid with simulated images to **verify** that the model would not fail even for these complicated patterns.

Hotspot Examples:



Etch Bias Model

Etch bias model, commonly referred to as **After Etch Intensity (AEI)** model, simulates the removal by etchant of unprotected materials.

Ideally, protected photoresist would not be removed at all during etching process. In reality, there is still erosion by the etchant, and the erosion rate may be **pattern dependent**. Commonly known pattern dependence including **loading effect**, i.e., higher density areas has lower erosion rate; **aspect ratio effect**, i.e., narrow trenches have slower erosion rate—both because etchant ions are more difficult to reach the resist patterns.

An AEI model predicts difference between resist profile before and after etch, i.e., **etch bias**. Etch bias is modeled as function of ADI patterns, where the function is defined as convolution of ADI patterns with some kernels. Also, etch bias model contains coefficient that are derived empirically.

ADI vs. ACI

Model Making	ADI	AEI
Image	Convolution of mask pattern with kernels	Convolution of ADI Contour (simulated or measured) with kernels
Wafer Data	Measurements of CD/image on resist pattern,	Measurements of CD/image difference between resist before and after etch.
Test pattern	Short range proximity	Short and middle range proximity
Kernel	Absorption, PEB, Development	Loading, Aspect Ratio

Resist/Etch Modeling on EDA Platform

As described before for optical modeling on an EDA platform, resist/etch modeling engine is not accessible to users. On a GUI for model building, users must select from available model forms, provide test pattern layouts and measurement data sheet in certain format(s) that the engine can read. User can also specify calibration spec, including numerical fitting method, accuracy, iteration, converging method, etc.

Each EDA company provides its own model forms. Model forms are fiercely protected Intellectual Property.

Chapter 3 Summary

In this chapter we discussed optical, resist, and etch models: what physics/chemistry are involved, what kind of model forms/functions are commonly available, how to create test patterns to calibrate models; how to build them using EDA tools.

It is worth to stress that modeling is among the most challenging, most laborious, and most competitive work in computational lithography. As we shall see in following chapters, corrections of mask patterns critically depends on accuracy and efficiency of models.

Chapter 3 Reference

Born: “*Principles of Optics*” by Max Born and Emil Wolf

Cobb: “Sum of Coherent Systems Decomposition by SVD” by Nick Cobb

Wong: “Optical Imaging in Projection Microlithography” by Alfred Kwok-Kit Wong

Mack: “The Lithographic Impact of Resist Model Parameters” by Mark Smith, Jeffrey Byers, Chris A. Mack, *Advances in Resist Technology and Processing XXI, SPIE Vol. 5376*, pp. 322-332

Dill: “Optical Lithography” by F.H. Dill, *IEEE Trans. Electron Devices*, ED-22, No. 7, pp. 440-444

Handbook: “Handbook of Microlithography, Micromaching, and Microfabrication: Edited by P. Rai-Choudhury, volume 1.